

Document: Test Cases and Result Report.

Test Cases Specification

Test case for STC89C52 (8051 Architecture) Simulator

Testing Framework: JUnit 5 (Java Automated Testing)

1. Automated Unit Test Cases (tests/)

All unit tests are organized under the tests/ directory and execute via JUnit 5 to independently validate each package (cpu, memory, instruction, and ui).

Test Case ID	Target File / Module	Test Scenario & Purpose	Input / Test Condition	Expected Output / State
TC-CPU-01	CPUTest.java	Immediate Data Transfer to ACC	MOV A, #50	ACC updated to 0x32 (50 ₁₀)
TC-CPU-02	CPUTest.java	Immediate Data Transfer to Register B	MOV B, #30	B updated to 0x1E (30 ₁₀)
TC-CPU-03	CPUTest.java	Addition & Auxiliary Carry Flag	ADD A, B (0x32 + 0x1E)	ACC = 0x50, AC = true, CY = false
TC-CPU-04	CPUTest.java	Carry Overflow on Maximum Value	ADD A, B (0xFF + 0x01)	ACC = 0x00, CY = true, AC = true, P = false
TC-CPU-05	CPUTest.java	Subtraction with Borrow	SUBB A, B (0x50 – 0x1E – 0)	ACC = 0x32, CY = false

Test Case ID	Target File / Module	Test Scenario & Purpose	Input / Test Condition	Expected Output / State
TC-CPU-06	CPUTest.java	Accumulator Increment	INC A (0x50 + 1)	ACC updated to 0x51
TC-CPU-07	CPUTest.java	Accumulator Decrement Underflow	DEC A (0x00 - 1)	ACC updated to 0xFF (wrap-around)
TC-CPU-08	CPUTest.java	Bitwise AND & Dynamic Parity Flag	ANL A, B (0x51 $\wedge$ 0x1E)	ACC = 0x10, P = true (odd parity: 1 bit set)
TC-CPU-09	CPUTest.java	Relative Short Jump Execution	SJMP 0x0005	PC updated to target address 0x0005
TC-CPU-10	CPUTest.java	Program Termination / Halt	END	CPU halted flag set to true
TC-MEM-01	MemoryTest.java	Program Memory File Loading	Text with ; comments & blanks	Comments and blank lines filtered out; clean code loaded
TC-STK-01	StackTest.java	Internal RAM Stack Operations	Push 0x55, then Pop byte	SP increments to 0x08, Pop returns 0x55,

Test Case ID	Target File / Module	Test Scenario & Purpose	Input / Test Condition	Expected Output / State
				SP resets to 0x07
TC-INS-01	InstructionTest.java	Assembly String Tokenization	Line: "MOV A, #50"	Mnemonic: "MOV", Op1: "A", Op2: "#50"
TC-SET-01	InstructionSetTest.java	Opcode Lookup & Validation	"MOV" vs "INVALID_OP"	"MOV" → true, "INVALID_OP" → false
TC-UI-01	UITest.java	Swing GUI Window Launch	Launch SimulatorUI	Swing GUI window instantiates without throwing exceptions

2. Integration Test Specification (programs/demo-program.txt)

This integration test executes a multi-step assembly program through the complete **FETCH** → **DECODE** → **EXECUTE** pipeline to verify that register transfers, arithmetic operations, and flag updates maintain correct state across sequential instructions

MOV A, #50 ; Load 50 (0x32) into ACC

MOV B, #30 ; Load 30 (0x1E) into B

ADD A, B ; ACC = 0x32 + 0x1E = 0x50, AC = true

INC A ; ACC = 0x50 + 1 = 0x51

ANL A, B ; ACC = 0x51 AND 0x1E = 0x10, P = true

END ; Stop pipeline

Test Results Report

1. Automated Unit Test Execution Results

All automated unit tests were executed using the **JUnit 5** framework. Every test passed with zero errors or failures.

Test Case ID	Target File / Module	Expected Result	Actual Output Result	Status
TC-CPU-01	CPUTest.java	ACC = 0x32	ACC = 0x32	PASS
TC-CPU-02	CPUTest.java	B = 0x1E	B = 0x1E	PASS
TC-CPU-03	CPUTest.java	ACC = 0x50, AC = true, CY = false	ACC = 0x50, AC = true, CY = false	PASS
TC-CPU-04	CPUTest.java	ACC = 0x00, CY = true, AC = true	ACC = 0x00, CY = true, AC = true	PASS
TC-CPU-05	CPUTest.java	ACC = 0x32, CY = false	ACC = 0x32, CY = false	PASS
TC-CPU-06	CPUTest.java	ACC = 0x51	ACC = 0x51	PASS
TC-CPU-07	CPUTest.java	ACC = 0xFF	ACC = 0xFF	PASS

Test Case ID	Target File / Module	Expected Result	Actual Output Result	Status
TC-CPU-08	CPUTest.java	ACC = 0x10, P = true	ACC = 0x10, P = true	PASS
TC-CPU-09	CPUTest.java	PC = 0x0005	PC = 0x0005	PASS
TC-CPU-10	CPUTest.java	halted = true	halted = true	PASS
TC-MEM-01	MemoryTest.java	Comments & blanks removed	Comments & blanks removed	PASS
TC-STK-01	StackTest.java	SP = 0x08 → Popped 0x55 → SP = 0x07	SP = 0x08 → Popped 0x55 → SP = 0x07	PASS
TC-INS-01	InstructionTest.java	Mnemonic: MOV, Op1: A, Op2: #50	Mnemonic: MOV, Op1: A, Op2: #50	PASS
TC-SET-01	InstructionSetTest.java	MOV → true, INVALID → false	MOV → true, INVALID → false	PASS
TC-UI-01	UITest.java	GUI window loaded without exception	GUI window loaded without exception	PASS

2. Integration Pipeline Trace Log Results (demo-program.txt)

The following trace log was generated during step-by-step execution of programs/demo-program.txt through the **FETCH** → **DECODE** → **EXECUTE** pipeline:

STEP 1: FETCH -> DECODE -> EXECUTE

- Executed Line : MOV A, #50
- Program Counter: 0x0000 -> 0x0001
- Register State : ACC: 0x32 | B: 0x00 | SP: 0x07 | DPTR: 0x0000
- Flags Computed : CY: false | AC: false | OV: false | P: false

STEP 2: FETCH -> DECODE -> EXECUTE

- Executed Line : MOV B, #30
- Program Counter: 0x0001 -> 0x0002
- Register State : ACC: 0x32 | B: 0x1E | SP: 0x07 | DPTR: 0x0000
- Flags Computed : CY: false | AC: false | OV: false | P: false

STEP 3: FETCH -> DECODE -> EXECUTE

- Executed Line : ADD A, B
- Program Counter: 0x0002 -> 0x0003
- Register State : ACC: 0x50 | B: 0x1E | SP: 0x07 | DPTR: 0x0000
- Flags Computed : CY: false | AC: true | OV: false | P: false

STEP 4: FETCH -> DECODE -> EXECUTE

- Executed Line : INC A
- Program Counter: 0x0003 -> 0x0004
- Register State : ACC: 0x51 | B: 0x1E | SP: 0x07 | DPTR: 0x0000
- Flags Computed : CY: false | AC: true | OV: false | P: false

STEP 5: FETCH -> DECODE -> EXECUTE

- Executed Line : ANL A, B

- Program Counter: 0x0004 -> 0x0005
- Register State : ACC: 0x10 | B: 0x1E | SP: 0x07 | DPTR: 0x0000
- Flags Computed : CY: false | AC: true | OV: false | P: true

STEP 6: FETCH -> DECODE -> EXECUTE

- Executed Line : END
- Execution State: [PROGRAM HALTED]

3. Final Register & Flag Output Verification

Comparison between expected end values and actual GUI Simulator display output after program execution:

Register / Flag	Expected Value	Actual Output	Verification Result
Program Counter (PC)	0x0005	0x0005	MATCH / PASS
Accumulator (ACC)	0x10	0x10	MATCH / PASS
Register B	0x1E	0x1E	MATCH / PASS
Stack Pointer (SP)	0x07	0x07	MATCH / PASS
Data Pointer (DPTR)	0x0000	0x0000	MATCH / PASS
Carry Flag (CY)	false	false	MATCH / PASS
Auxiliary Carry Flag (AC)	true	true	MATCH / PASS
Overflow Flag (OV)	false	false	MATCH / PASS
Parity Flag (P)	true	true	MATCH / PASS

Status: Passed